Naman Saxena

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Staff Design Verification Engineer with over eight years of experience across IP-, subsystem-, SoC-, and CPU-core-level verification. Currently verifies advanced ARM-architecture CPU cores for Snapdragon processors at Qualcomm, focused on Memory Management Units, Translation Lookaside Buffers, and prefetchers. Background spans custom CPU DV, GPU DV, memory and storage verification IP development, and academic research in computer architecture and CMOS circuit design.

Skills

Languages & Scripting:	Python, C, C++, CUDA, Perl, Bash, TCL, Assembly
HDL & Verification Methodology:	SystemVerilog, Verilog, UVM, OVM, Formal Verification, Functional Coverage
EDA & Debug Tools:	Questa Sim, Synopsys VCS, Cadence Xcelium, Vivado, Verdi, Indago, DVE
Version Control & Collaboration:	Git, Perforce, ClearCase, JIRA, Confluence, GitLab, Notion
Operating Systems:	Linux/Unix, Windows

Experience

Qualcomm — Staff Engineer, Custom CPU Design Verification *Dec 2024 – Present | Santa Clara, CA*

- Verify advanced ARM-architecture CPU cores for Snapdragon processors used across 85+ PC models and millions of smartphones, laptops, and automotive platforms.
- Lead verification of proprietary micro-architectural features that enhance Memory Management Unit (MMU) performance; verify integration of the ARM Realm Management Extension.
- Own the MMU/TLB invalidation checker end-to-end, from feature development through maintenance; found approximately 30 critical bugs and revamped its coverage model for comprehensiveness and readability.
- Serve as verification point-of-contact for all CPU core prefetchers; run cross-team knowledge-transfer sessions.

Qualcomm — Senior Engineer, Custom CPU Design Verification *Feb 2023 – Dec 2024 | Santa Clara, CA*

- Led verification of all data, instruction, and translation prefetchers across multiple CPU projects, uncovering 100+ design issues.
- Spearheaded accelerated verification of a TLB hardware-trace feature with cross-functional teams, identifying roughly 10 design issues, including a previously undetected cross-generational issue.
- Rapidly built MMU expertise; found approximately 25 bugs while maintaining design regression health.

Apple — GPU Design Verification Summer Intern (SEG-GPU) *May 2022 – Aug 2022 | Orlando, FL*

- Verified an MMU bus functional model (BFM) using SystemVerilog and UVM.
- Built a functional DV-level performance model for the MMU and streamlined the regression flow, identifying 3 performance bugs early in the verification cycle.

Qualcomm — Senior DV Engineer, CPU Subsystem *Sep 2020 – Aug 2021 | Bengaluru, India*

- Owned verification of two new features (dynamic memory and core-frequency control) from scratch, including a test plan with 28 directed sequences and 25 SystemVerilog assertions on a UVM-based setup.
- Collaborated across Design, SoC, and Memory teams; found 2 critical SoC bugs and presented findings to 100+ engineers.
- Improved regression runtime by approximately 15% by eliminating redundant tests and script inclusions.

Mentor Graphics (Siemens EDA) — Senior Member Technical Staff, DRAM Verification IP *Jan 2019 – Sep 2020 | Noida, India*

- Developed a standalone verification IP for the DDR5 Memory & DIMM sideband interface, including the BFM and a UVM-based test-bench with sequences, assertions, and a logger component.
- Reconstructed the callback architecture, configuration, and backdoor APIs across all DRAM verification IPs for scalability.
- Recognized with a certificate of excellence and spot bonus for a Python-based DRAM VIP configuration framework.

Mentor Graphics — Member Technical Staff, Storage Verification IP *Jun 2017 – Dec 2018 | Noida, India*

- Built a complete SATA v3.3 compliance test suite (~100 directed scenarios) in SystemVerilog.
- Implemented the PIPE interface BFM (Spec v4.4.1) from scratch.
- Recognized with outstanding feedback for customer engagement on the SATA verification IP.

Education

University of Michigan, Ann Arbor — M.S., Electrical Engineering & Computer Science *Aug 2021 – Dec 2022*

GPA: 4.0 / 4.0

Coursework: Parallel Computer Architecture, Computer Architecture, Applied Parallel Programming with GPUs, Deep Learning for Computer Vision, Advanced Compilers, VLSI Design I, Algorithms

Delhi Technological University — B.Tech., Electrical & Electronics Engineering *Aug 2013 – May 2017*

CGPA: 9.1 / 10.0 — Branch Topper (3rd rank), Meritorious Student Award, 3rd & 4th year

Coursework: Computer Architecture, Microprocessors & Applications, Linear Integrated Circuits, Digital Circuits & Systems, Digital Signal Processing, Microcontrollers & Embedded Systems

Selected Publications

- N. Saxena, N. Goel, and R. Rastogi, “SATA Specification 3.3 Gaps Filled by SATA QVIP,” *Verification Horizon*, Vol. 14, Issue 1, Mentor – A Siemens Business, 2018.
- N. Saxena, S. Dutta, N. Pandey, and K. Gupta, “Implementation and Performance Comparison of a Four-Bit Ripple Carry Adder Using Different MOS Current Mode Logic Topologies,” *2017 International Conference on Computational Science and Its Applications (ICCSA)*, Springer.
- 4 additional publications and poster presentations (2016–2018) in peer-reviewed journals and IEEE/Springer conference proceedings.

Awards & Recognition

- Promoted from Senior Engineer to Staff Engineer, Qualcomm (2024)
- First Runner-Up & People’s Choice Award, NASA Space Apps Challenge (2015)
- Best Project Award, EECS 598 (Applied Parallel Programming with GPUs), University of Michigan (2021)
- Certificate of Excellence & Spot Bonus, Mentor Graphics (2019)
- Branch Topper / Meritorious Student Award, Delhi Technological University (2017)

Research Experience

- **Visiting Research Scholar, Loughborough University, UK** (2016) — Prototyped a point-of-care E. coli detection method using a novel opto-acoustic transduction technique; awarded the Wolfson Summer Bursary (£2,000).
- **Research Assistant, Delhi Technological University** (2015–2017) — Researched CMOS technologies and advanced memory-cell designs; co-authored 4 publications.
- **Research Internship, IIT Bombay** (2015) — Developed a microcontroller-based web server for remote DC-motor control and a closed-loop buck converter for virtual-lab applications.

Professional Service & Leadership

- Peer reviewer for 10+ IEEE/Springer conferences and journals (~50 manuscripts reviewed in 2025), including SOCC, ICCD, ICCCNT, EEEIC, and JETTA.
- Technical Program / Program Committee Member: ICCD, SOCC, SBCCI, and ISCI (2025).
- Mentor, Startupbootcamp accelerator (2025); Judge, Technovation Girls global competition (2025).
- Grader, University of Michigan — Computer Architecture, Digital Integrated Circuits, Micro/Nano Device Fabrication, Introduction to Semiconductor Devices (2022).
- Tutor, Chegg.com — 300+ live tutoring sessions for university-level students, 90% positive rating (2013–2017).